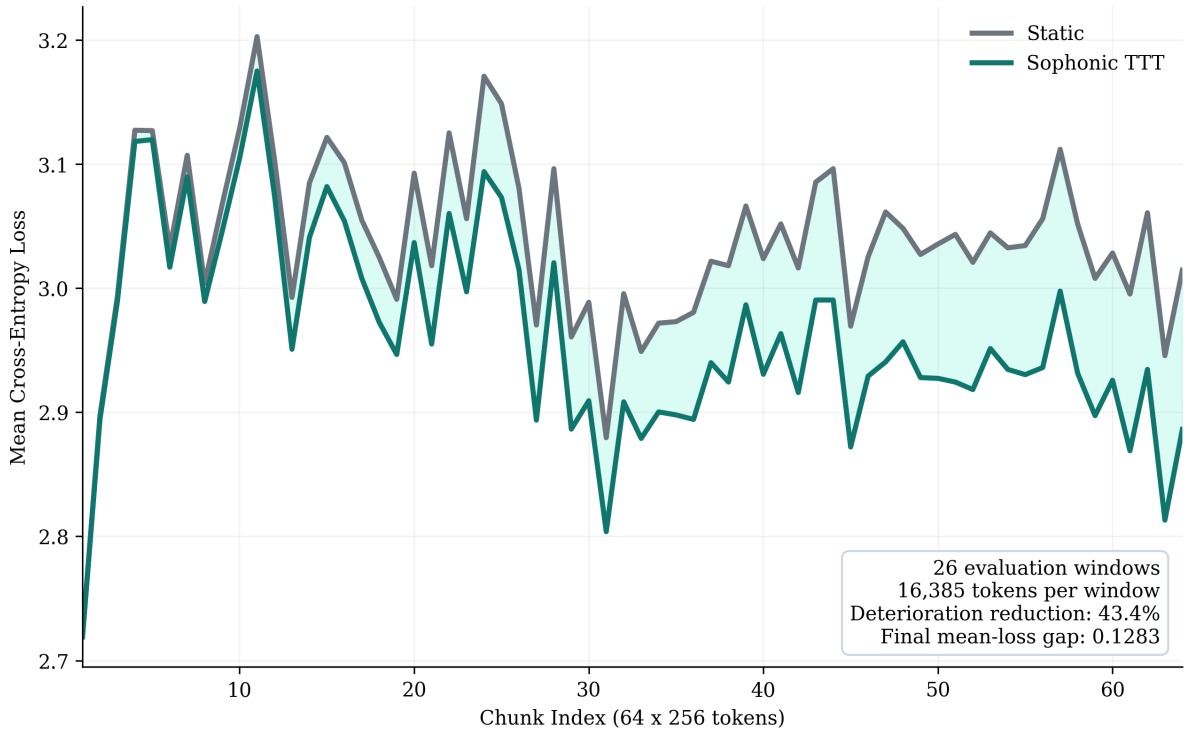

Figure 4: Long-horizon continual adaptation on Qwen2.5-1.5B and PG-19. Averaged over 26 A40 evaluation windows of length 16,385 tokens, Sophonic TTT remains below static throughout the document and reduces overall deterioration by 43.4%.

9 Experiment 6: Continual Test-Time Adaptation

The same frozen-base plus late-delta pattern can also be used online. Instead of training Sophonic modules once and keeping them fixed, we attach late-layer LoRA-style factors to a frozen base model and update only those factors as a document unfolds. This turns the adapter into an explicit state object. It can be reset, carried across documents, saved, reloaded, and compared as an artifact in its own right. This is not merely a cheaper version of full-model test-time training. It has properties that full TTT fundamentally cannot have, because the base weights remain untouched.

We evaluated continual adaptation on Qwen2.5-1.5B with PG-19 windows using a chunked left-to-right protocol. Each document window is split into fixed-size chunks, loss is measured on every chunk, and the adaptive method receives one gradient step after each chunk except the last. The default Sophonic continual configuration targets the last 6 layers, covers both MLP and attention matrices, and uses rank-8 factors. GPU references were run on an A40 in `bfloat16`; the same positive trend also appeared locally on an M2 MacBook Air, where full-model TTT was not feasible.

9.1 Long-Horizon Continual Adaptation

The large-scale continual result is straightforward. On 26 PG-19 evaluation windows of length 16,385 tokens (64 chunks of 256 tokens), static evaluation deteriorates from 2.7191 to 3.0144 mean loss ($\Delta = +0.2954$). Sophonic TTT deteriorates from 2.7191 to 2.8861 ($\Delta = +0.1671$). That is a 43.4% reduction in document-level deterioration relative to static evaluation. The effect also strengthens with horizon: the corresponding matched local result at the same 64×256 window length was 23.6%, and shorter 16×256 and 32×256 pilots were weaker still.